Enhancing Transaction Monitoring With Machine Learning Model

Making Smarter Decisions To Detect Suspicious Transactions Beyond Traditional Rule-Based Systems

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PROBLEM STATEMENT USE CASE

Transaction #	Month	OCR	Amount	Suspicious?
TX001	JAN	11111	1200	✗
TX002	APR	112222	10600	✗
TX003	APR	2211111	680	✗
TX004	JUL		9260	✓
TX005	NOV	1000001	11050	✗

Problem Statement $\xrightarrow[\text{through ML Model}]{\text{Automate}}$ **Solution**

Problem Statement
- Transactions are manually inspected
- Man hours are lost
- Prone to Human Error

Automate through ML Model

Solution
+ Build an ML Model
+ Man Hours are saved
+ Speed (faster detection)

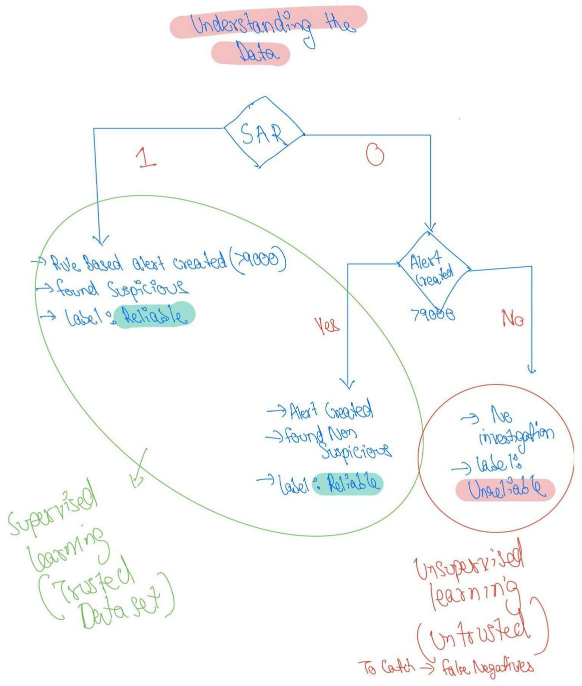
GIVEN DATA AND APPROACH

1. Reliable Dataset (Above 9,000)

- **Definition**: Transactions exceeding **9,000** (flagged by the rule-based system).
- **Characteristics**:
  - Includes reliable SAR labels (investigated cases).
  - Suitable for **supervised learning** to train the model.

2. Unreliable Dataset (Below 9,000)

- **Definition**: Transactions below **9,000** (neither flagged or investigated).
- **Characteristics**:
  - Non Reliable SAR labels.
  - Suitable for **unsupervised learning** to detect anomalies.



Researching about the Data

Amid rampant fraud and abuse targeting older adults, FinCEN urges financial institutions to detect, prevent, and report suspicious financial transactions.

Elder financial exploitation (EFE) is defined as the illegal or improper use of an older adult's funds, property, or assets.[1]

SAR Filing Request:
FinCEN requests that financial institutions reference the advisory by including "**EFE FIN-2022-A002**" in SAR field 2 ("Filing Institution Note to FinCEN"), and mark the check box for elder financial exploitation.

Introduction

The Financial Crimes Enforcement Network (FinCEN) is issuing this advisory to alert financial institutions to the rising trend of EFE targeting older adults[2] and to highlight new EFE typologies and red flags since FinCEN issued the first EFE Advisory in 2011.[3] FinCEN is also issuing this advisory in support of World Elder Abuse Awareness Day, which has been commemorated on June 15 every year since 2006 and provides an opportunity for communities around the world to promote a better understanding of abuse and neglect of older adults by raising awareness of the related cultural, social, economic, and demographic factors.[4]

According to the U.S. Department of Justice, elder abuse, which includes EFE among other forms of abuse, affects at least 10 percent of older adults each year in the United States,[5] with millions of older adults losing more than $3 billion to financial fraud annually as of 2019.[6] Despite the

Common Patterns of Suspicious Activity

Some of the common patterns of suspicious activity identified by the Financial Crimes Enforcement Network are as follows:[1]

- A lack of evidence of legitimate business activity (or any business operations at all) undertaken by many of the parties to the transactions(s)
- Unusual financial nexuses and transactions occurring among certain business types (for example, a food importer dealing with an auto parts exporter)
- Transactions not commensurate with the stated business type or that are unusual compared with volumes of similar businesses operating locally
- Unusually large numbers and/or volumes of wire transfers, repetitive wire transfer patterns
- Unusually complex series of transactions involving multiple accounts, banks, and parties
- Bulk cash and monetary instrument transactions
- Unusual mixed deposits into a business account
- Bursts of transactions within short periods, especially in dormant accounts
- Transactions or volumes of activity inconsistent with the expected purpose of the account or activity level as mentioned by the account holder when opening the account
- Transactions attempting to avoid reporting and recordkeeping requirements.

Money laundering

Money laundering indicators

A customer:

- who holds a visitor or other short-term visa, opens an account during a short stay in Australia
- is suspected of using a personal account for business purposes, or vice-versa
- uses large amounts of cash to pay for international funds transfers
- makes frequent cash deposits and/or withdrawals over multiple days
- has unusually large volumes of cash deposits and withdrawals, inconsistent with their profile
- has third parties who structure cash deposits into their bank account
- has third-party transfers to and from their account with no apparent economic rationale
- has transactions that are inconsistent with their profile and/or transaction history
- makes a rounded-sum transaction that appears inconsistent with what is expected from the customer
- appears to use their accounts to transfer money through to other accounts only, in an attempt to layer illicit funds
- has an inactive account that begins to see frequent financial activity
- makes transactions that are unnecessarily complex for their declared purpose
- makes transactions that involve persons or entities identified by media, law enforcement and/or intelligence agencies as being linked to criminal activities
- uses complex company and/or trust structures and associated banking arrangements in an attempt to obscure the source and beneficial ownership of their funds
- opens accounts using front or shell companies
- purchases high-value assets that are inconsistent with their profile
- attempts to layer funds through both domestic and international funds transfers to accounts held with multiple domestic and foreign banks
- splits large cash deposits into multiple smaller deposits below the $10,000 reporting threshold (this is known as structuring)
- makes multiple transactions when one transaction would suffice

Sources :

- https://www.unit21.ai/blog/financial-and-bank-suspicious-activity-examples
- Indicators of suspicious activity for the banking sector | AUSTRAC
- (PDF) A Survey of Machine Learning Based Anti-Money Laundering Solutions
- :https://www.austrac.gov.au/business/how-comply-guidance-and-resources/guidance-resources/indicators-suspicious-activity-banking-sector-0

FinCEN Issues Analysis on Elder Financial Exploitation

Immediate Release: April 18, 2024

Financial Institutions Report $27 Billion in Elder Financial Exploitation Suspicious Activity in One-Year Period

WASHINGTON—The U.S. Department of the Treasury's Financial Crimes Enforcement Network (FinCEN) issued a Financial Trend Analysis today focusing on patt
Bank Secrecy Act (BSA) data linked to Elder Financial Exploitation (EFE), or the illegal or improper use of an older adult's funds, property, or assets. FinCEN exan
between June 15, 2022 and June 15, 2023 that either used the key term referenced in FinCEN's June 2022 EFE Advisory or checked "Elder Financial Exploitation
This amounted to 155,415 filings over this period indicating roughly $27 billion in EFE-related suspicious activity.

mary objective of AML works is to identify the most approaches to recognize suspicious activities with the ccuracy and performance measurements possible. In chieve these goals, it is critical to execute proper data n, data transformation, feature selection, and data In this survey, we explore all published research conduct an in-depth evaluation in terms of ML-based money laundering detection systems. To the best of our knowledge, we have covered all recently published papers.

Based on our findings, class imbalance and the substantial volume of transaction data are common challenges found in most research works, yet only a single paper [124] has recognized or partially addressed this issue. Moreover, some papers [67][111][96][114] suggest to apply semi-supervised techniques as a more suitable approach to handle a significantly large volume of data.

5. Wire Transfer Fraud

This is a broad term used to describe any situation in which malicious activity occurs during the course of a **wire transfer**. Perhaps one of the most well-known wire transfer scams was the "Nigerian Prince" email fraud, which happened in the early 2000s (**and still makes a considerable amount of money today**).

DATA OVERVIEW

Transactions Dataset

1. **`Customer_ID`**: Unique identifier for the customer.
2. **`Customer_ID_Counterparty`**: Identifier for the counterparty involved in the transaction (if applicable).
3. **`Transaction_Value`**: Monetary value of the transaction (positive for deposits, negative for withdrawals).
4. **`Transaction_Type`**: Type of transaction, such as wire transfer, card etc.
5. **`Month`**: The month when the transaction occurred.

Labels Dataset

1. **`Customer_ID`**: Unique identifier for the customer (matches Customer_ID in the transactions dataset).
2. **`Month`**: The month corresponding to the transactions.
3. **SAR**: Indicates if a Suspicious Activity Report (SAR) was filed for the customer **for that month** (1 = Yes, 0 = No).

KYC Dataset

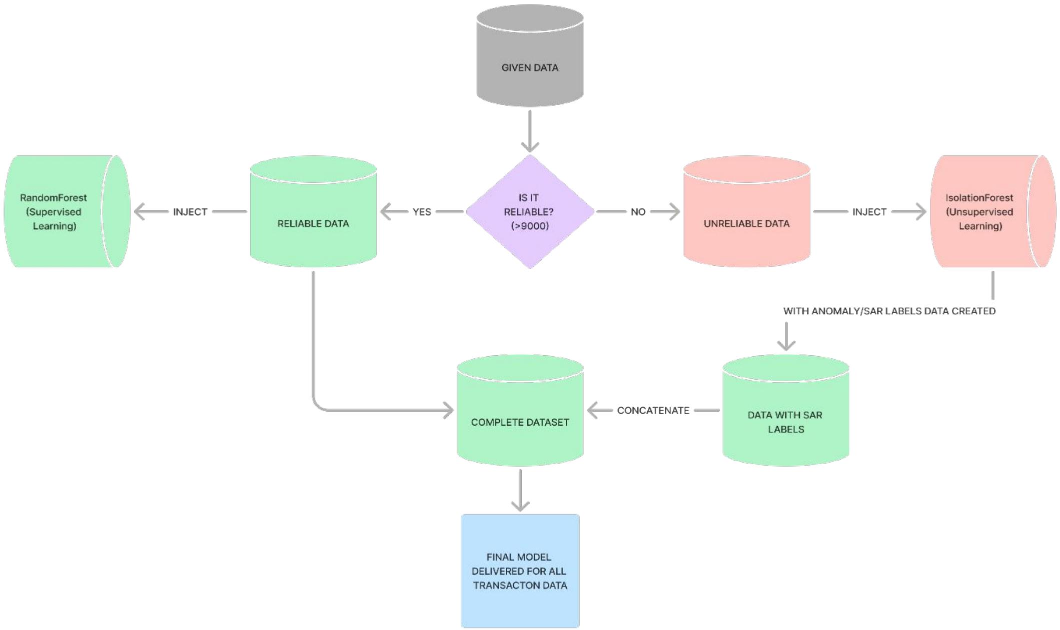
1. **`Customer_ID`**: Unique identifier for the customer (matches Customer_ID in the transactions dataset).
2. **`Age`**: Age of the customer.
3. **`Sex`**: Gender of the customer (e.g., Male/Female).
4. **`High_Risk_Country`**: Boolean flag indicating if the customer is associated with a high-risk country (True = Yes, False = No).
5. **`Vulnerable_Area`**: Boolean flag indicating if the customer operates in a vulnerable region.
6. **`Intention_International_Payments`**: Boolean flag indicating if the customer had an intent to make international payments.
7. **`Intention_Cash_Deposits`**: Boolean flag indicating if the customer had an intent to make significant cash deposits.

TOOLS AND TECHNIQUES

- **Tools**:
    - Figma - for intuitive diagrams
    - Notability - notes
    - Python (Pandas, Scikit-learn, Matplotlib/Seaborn).
    - Jupyter Notebook (Gemini Integrated - For utility scripts)
    - Libraries for anomaly detection and machine learning (RandomForest, Isolation Forest).
    - GitHub - for commit and remote repository
    - ChatGPT - for redirected towards the research papers
    - ResearchGate - For research papers
    - Google Slides - for presentation
    - Google Sheets - for exploring the datasets

- **Techniques**:
    - Supervised and unsupervised learning.
    - Data preprocessing and feature engineering.

Machine Learning Approach + Code Walkthrough



Prediction and Results

6. Predictions and Results

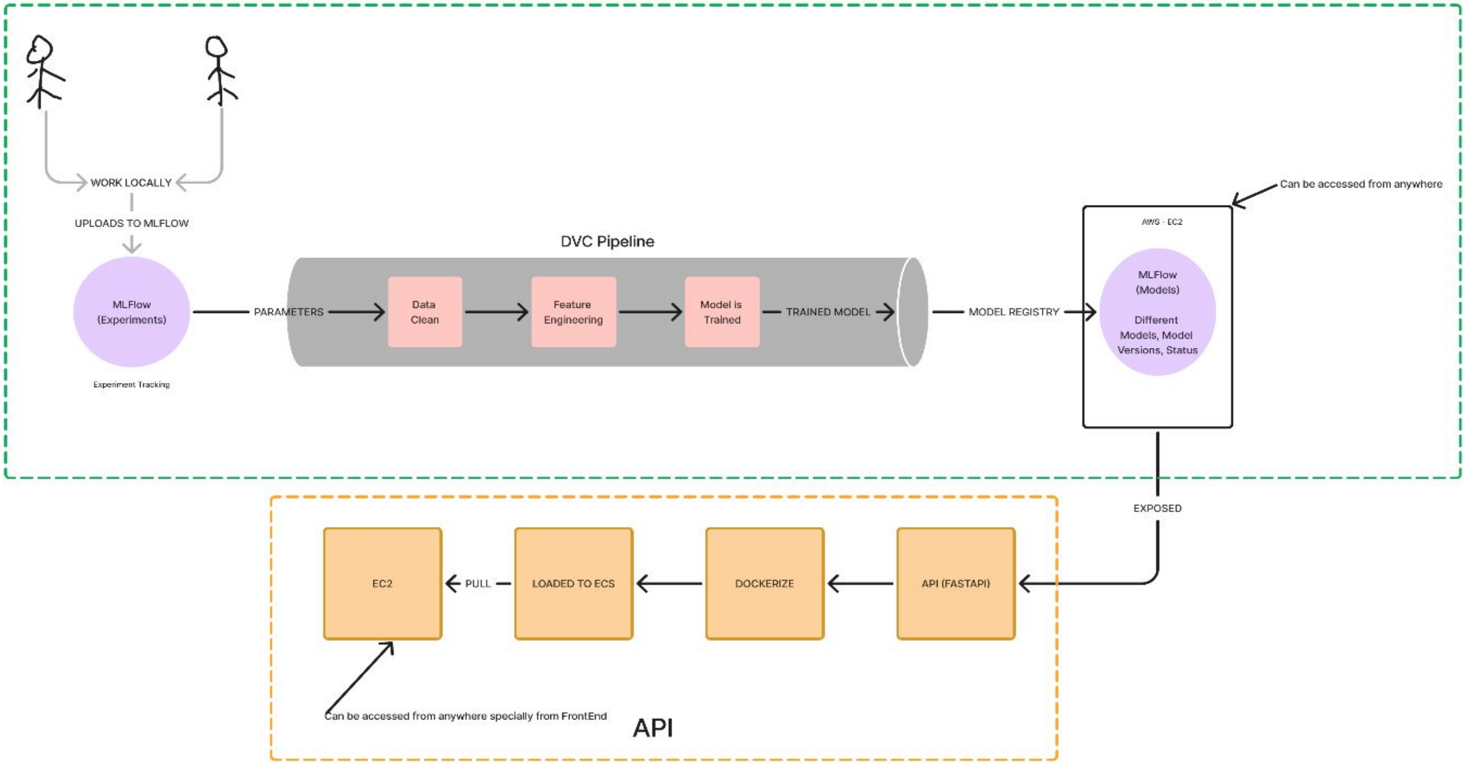
- **Predictions**:
    - High-risk customers flagged by the model.
    - Detection below the 9,000 threshold.
- **Results**:
    - Precision, recall, and F1-score metrics.

```
Accuracy: 0.94
ROC-AUC Score: 0.96
Classification Report:
              precision    recall  f1-score   support

           0       0.95      0.98      0.97      2111
           1       0.84      0.63      0.72       289

    accuracy                           0.94      2400
   macro avg       0.89      0.81      0.84      2400
weighted avg       0.94      0.94      0.94      2400
```

Conclusion

- **Conclusion**:
  - Summary of key findings.
  - Age plays a major factor in the given dataset.
  - With more time, the confidence on the model can be improved by hyper parameter tunings and more feature engineering

NEXT STEPS


WORK LOCALLY
UPLOADS TO MLFLOW
MLFlow (Experiments)
Experiment Tracking
DVC Pipeline
PARAMETERS
Data Clean
Feature Engineering
Model is Trained
TRAINED MODEL
MODEL REGISTRY
Can be accessed from anywhere
AWS - EC2
MLFlow (Models)
Different Models, Model Versions, Status
EXPOSED
API (FASTAPI)
DOCKERIZE
LOADED TO ECS
PULL
EC2
Can be accessed from anywhere specially from FrontEnd
API

THANK YOU